SPACE MECHANICS

PROFESSIONAL ELECTIVE – V

VII Semester									
Course Code	Category	Hours / Week			Credits	Maximum Marks		
		L	T	P	C	CIE	SEE	Total
A6AE55	PCC	3	0	0	3	40	60	100

UNIT-I BASIC CONCEPTS AND THE GENERAL N-BODY PROBLEM

BASIC CONCEPTS: The solar system, Reference frames and coordinate systems, The celestial sphere, The ecliptic, Motion of vernal equinox, Sidereal time, Solar Time, Standard Time, The earth's atmosphere
THE GENERAL N-BODY PROBLEM: The many body problem, Lagrange-Jacobi identity. The circular restricted three-body problem, Libration points, Relative Motion in the N-body problem

UNIT-II THE TWO-BODY PROBLEM ANDTHE LAUNCHING OF A SATELLITE

THE TWO-BODY PROBLEM: Equations of motion-General characteristics of motion for different orbits-Relations between position and time for different orbits, Expansions in elliptic motion, Orbital Elements. Relation between orbital elements and position and velocity.

THE LAUNCHING OF A SATELLITE: Launch vehicle ascent trajectories, General aspects of satellite injection. Dependence of orbital parameters on in-plane injection parameters, Launch vehicle performances, Orbit deviations due to injection errors

UNIT-III PERTURBED SATELLITE ORBITS AND INTERPLANETARY TRAJECTORIES

PERTURBED SATELLITE ORBITS: Special and general perturbations- Cowell's Method, Encke's method. Method of variations of orbital elements, General perturbations approach
INTERPLANETARY TRAJECTORIES: Two-dimensional interplanetary trajectories, Fast interplanetary trajectories, Three dimensional interplanetary trajectories. Launch of interplanetary spacecraft. Trajectory about the target planet

UNIT-IV BALLISTIC MISSILE TRAJECTORIES

The boost phase, the ballistic phase, Trajectory geometry, optimal flights. Time offlight, Re-entry phase. The position of the impact point, Influence coefficients

UNIT-V LOW-THRUST TRAJECTORIES

Equations of Motion. Constant radial thrust acceleration, Constant tangential thrust(Characteristics of the motion>), Linearization of the equations of motion, Performance analysis.

Text Books:

1. William E. Wiesel (2010), *Spaceflight Dynamics*, 3rd edition, McGraw-Hill, New Delhi.
2. JJ Sellers," Understanding the Space", CEI publication.

Reference Books:

1. *Vladimir A. Chobotov (2002*), Orbital Mechanics, 3rd Edition, AIAA Education Series, USA.
2. *David A. Vellado (*2007), Fundamentals of Astrodynamics and Applications, 3rd Edition, Springer, Germany.
3. *J. W. Cornelisse* (1979), Rocket Propulsion and Spaceflight Dynamics, Pitman Publishing, London.

COURSE OUTCOMES : STUDENTS SHOULD BE ABLE TO:

1. Explain fundamental components related to the solar system and their implications in celestial mechanics
2. Evaluate aspects related to the launching of satellites, including launch vehicle ascent trajectories, satellite injection, and the impact of injection errors on orbital parameters and launch vehicle performances.
3. Examine interplanetary trajectories, including two-dimensional and three-dimensional trajectories.
4. Evaluate ballistic missile trajectories.
5. Analyze characteristics of motion with constant radial and tangential thrust